Wöden Kusner

Applied mathematician | certified computation, geometry, and optimization
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Summary

Mathematician (Ph.D., University of Pittsburgh, advisor Thomas C. Hales) with twelve years of research in discrete geometry, optimization, and computer-assisted proof. Builds rigorous numerical software (interval arithmetic, branch-and-bound search, independent certificate checking) and turns hard technical problems into verified results. Experienced explaining mathematics to non-specialists, mentoring, and running a campus-wide quantitative support program.

Technical Skills

- **Languages and tools:** Mathematica and Python (main tools; incl. mpmath); working familiarity with C (FLINT/Arb), Julia, and HTML/CSS/JavaScript; Git/GitHub; LaTeX.
- **Methods:** interval arithmetic and certified computation (FLINT/Arb); branch-and-bound search; nonlinear optimization and critical-point analysis; computational geometry; discrepancy and quasi-Monte Carlo point sets; probability and statistics.
- **Workflow:** AI-assisted research and software development (Claude, Claude Code); reproducible releases with archived DOIs.

Selected Projects

- **kisscrit** (2026, open source). C library on FLINT/Arb for certified search of critical configurations of kissing spheres: interval branch-and-bound plus a separate certificate checker. Released v0.1.0 with DOI 10.5281/zenodo.23603268.
- **Spherical cap discrepancy** (2 papers in preparation; code and data public). Exact optimal configurations of $N \leq 6$ points on the sphere, with a computer-assisted proof for $N = 5, 6$ cross-verified by two independent interval-arithmetic pipelines (Arb and mpmath.iv); open-source `capdisc` verification code, certificate checker, and an atlas of critical configurations for $N = 5$–12 (github.com/wkusner/discrepancy).
- **Critical radii of kissing configurations** (in preparation). Algebraicity and balance results for twelve-sphere configurations, certified with kisscrit.
- **Computer-assisted packing proofs.** Optimality of packings of regular pentagons, with T. Hales (in revision, *Trans. Amer. Math. Soc.*); local optimality of the double lattice packing, with Y. Kallus (*Discrete Comput. Geom.*, 2016); density bounds for cylinders and polycylinders (*Discrete Comput. Geom.*, 2014, 2016).
- **Chirality measure for curves** (with G. Dietler, E. Rawdon, R. Kusner, P. Szymczak; in revision). Tensor-valued chirality measure built from hydrodynamic intuition; applies to polygonal curves, polymers, and biomolecules.
- **Hyperuniform point sets on the sphere** (with J. Brauchart, P. Grabner, J. Ziefle; *Constr. Approx.* 2019, *Monatsh. Math.* 2020). Deterministic and random constructions of well-distributed point sets.
- **Of Instruments** (2026). Interactive browser versions of historical mathematical instruments (slide rules, planimeters, Napier's bones, verniers), written in JavaScript.
- **Community data infrastructure.** Fox Valley Data Exchange data-lakehouse specification working group (2025); Community Data Hub data team for Imagine Fox Cities (2021–2022).

Experience

Director, Quantitative Resource Center; Adjunct Assistant Professor of Mathematics
2025–present
Lawrence University

- Run quantitative peer tutoring for the college: recruit, train, and mentor peer tutors; coordinate support with departments; advise on ALEKS math placement.
- Created the Quantitative Learning Assistant program (2025): wrote the position description, hire and schedule assistants, and pair each with a course instructor.
- Provide individualized support for case-managed students and hold open office hours for students, faculty, and staff.

Instructor of Mathematics 2023–2025
Lawrence University

- Designed the ALEKS-based course structure and exercise pack for Preparation for Calculus, later used by colleagues teaching the course; supervised directed studies and summer undergraduate research.

Assistant Professor (non-tenure-track) 2020–2022
University of Georgia

- Taught Calculus I and II; UGA Teaching Academy Fellow; member of the instructional corps recognized by the 2022 Regents' Teaching Excellence Award.

Assistant Professor (non-tenure-track) and Postdoctoral Associate 2017–2020
Vanderbilt University

- Research in the Center for Constructive Approximation; Data Science Institute faculty affiliate.
- Taught probability, statistical inference, and multivariable calculus; supervised teams of teaching assistants and graders; mentored undergraduate research.
- Organized the Computational Analysis Seminar; co-organized the Shanks Workshop on Computations and Linear Programming Bounds for Energy, Packing and Covering (2019).

FWF Postdoctoral Researcher 2014–2017
Graz University of Technology, Austria

- Research on point configurations, discrepancy, and packing; taught a course on packings, lattices, and configurations.

Education

Ph.D., Mathematics, University of Pittsburgh 2014
Dissertation: *Bounds on packing density via slicing* (advisor Thomas C. Hales)
B.S., Mathematics, Haverford College 2007

Publications, Talks, and Professional Activity

- 7 peer-reviewed papers (incl. 3 in *Discrete & Computational Geometry*) and 6 papers in revision or preparation; full list at wkusner.github.io.
- 37 invited and contributed talks, including Oberwolfach, MSRI, ICERM, AIM, BIRS–CMO, and the Isaac Newton Institute.
- Referee for *Annals of Mathematics*, *Discrete & Computational Geometry*, IMRN, *SIAM J. Discrete Math.*, and others; NSF Graduate Research Fellowship reviewer (2022).
- Languages: English (native), German (intermediate), Spanish (elementary).